A-A-56017
12 April 1990

COMMERCIAL ITEM DESCRIPTION

CELLULOSE ACETATE

The General Services Administration has authorized the use of this Commercial Item Description.

Abstract. This Commercial Item Description describes the requirements for four grades of one type of cellulose acetate used as an ingredient in solid rocket propellants.

Salient characteristics.

1. The nominal composition of the cellulose acetate shall consist essentially of the chemicals acetyl and combined acetic acid containing up to 3 percent moisture, with no more than trace amounts of free acidity as acetic acid and with only trace amounts of chlorine and ash.

2. The viscosities of and minimum/maximum allowable percentages of ingredients and impurities for each grade are listed in table I, along with required characteristics for the material and a listing of test documents used to determine these characteristics.

3. The cellulose acetate shall be manufactured under conditions and procedures standard to the industry, and shall be supplied as a dry powder that is uniform in quality and free of foreign matter.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Missile Command ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5207.

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TABLE I. Required characteristics and test documents.

Characteristics	Grade A		Grade B		Grade C		Grade D		Test Document (ASTM)
	Min	Max	Min	Max	Min	Max	Min	Max	
Viscosity, seconds	1.8	3.9	6.0	13.0	53	79	22	38	D-1343
Free Acidity as Acetic Acid, percent		0.1		0.1		0.1		0.1	D-871
Average Acetyl Content, percent	39.3	40.3	39.3	40.3	38.9	39.9	38.9	39.9	D-871
Average Combined Acetic Acid, percent	55	56	54.8	56.2	54.5	55.5	53.7	55.7	D-871
Moisture, percent		3.0		3.0		3.0		3.0	D-871
Chlorine, percent		0.2		0.2		0.2		0.2	D-2641
Ash, percent		0.2		0.2		0.2		0.2	D-871

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. None.

Preservation, packaging, packing, labeling, and marking. The cellulose acetate shall be packed in 50-pound or 25-kilogram multiwall bags, as directed by the contract. Containers shall comply with common carrier regulations applicable to the mode of transportation selected.

Notes.

(a) WARNING! Handlers of cellulose acetate should be thoroughly familiar with all safety related aspects of the material. Like most organic materials in powder form, cellulose acetate dust is potentially explosive. This aspect of safety is described in NFPA Pamphlet No. 654. Suppliers should be required to supply full safety protective information relating to their products.

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(b) Referenced document source. Application for copies of ASTMs referenced in this document should be addressed to American Society of Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187. Copies of NFPA Pamphlet No. 654 can be obtained from the National Fire Protection Association, Battery March Park, Quincy, MA 02269.

ASTM D871	-	Standard Methods of Testing Cellulose Acetate
ASTM D1343	-	Standard Test Method for Viscosity of Cellulose Derivatives by Ball-Drop Method
ASTM D2641	-	Standard Test Method for Chlorine in Cellulose
NFPA Pamphlet No. 654	-	Prevention of Fire and Dust Explosions in the Chemical, Dye, Pharmaceutical, and Plastics Industries

(c) Other notes. Products conforming to grades A and B are manufactured by Eastman Chemical Products, Inc. (CAGEC 74364), and identified as CA-398-3 and CA-398-6, respectively. A product identified as CA-394-60 SLF closely matches desired parameters for grade C, but is an unfiltered powder. Users should carefully evaluate manufacturer's specifications for suitability of material for their particular purposes.

MILITARY INTERESTS:

Custodians

Army - MI

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA

PREPARING ACTIVITY:

Army - MI

Project 6810-A044

FSC 6810

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